

Zinger Polymarket Trading Bot

Session Progress, Current State & Improvement Roadmap

Compiled from DEVELOPMENT.md, live observability API, and operator session

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Abstract

This report documents work completed on the **Zinger** paper-trading bot for Polymarket BTC/ETH 5-minute up/down markets between late August and 2 September 2026. It covers infrastructure delivered (arb engine, directional stack, observability, backtest), the current live paper session, P&L attribution, known failure modes, and a prioritized improvement plan. Data snapshot: session `session-mtjqc931`, equity **\$884.51** (−11.5% from \$1,000 reset), 45 session closes, 13% directional win rate on 100-trade lookback, arb packages net positive at package level.

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Executive Summary

What Zinger Is

Zinger is a TypeScript full-stack system (`/home/david/Zinger`) comprising:

- A **scan loop** on Polymarket CLOB/Gamma for BTC and ETH 5m binaries
- Two engines: **directional** (TA + book + optional strike forecast) and **atomic arb** (buy both legs when $UP+DOWN < \$1$ minus fees)
- **Paper and live** modes with operator dashboard (port 3000)
- **Governor** regime switching, edge gate, signal health, and data assurance layers

Bottom Line

Area	Assessment
Infrastructure & reliability	Strong — observability, arb rollback, settlement fixes, config authority
Arb strategy (package level)	Working — 2 packages, +\$0.86 to +\$0.68 net; rare on live books
Directional edge (paper)	Weak — session directional P&L roughly $-\$15$ on 57 closes
6-month synthetic backtest	Marginal — +\$60 on \$10k arb-only (+0.6%), synthetic CLOB
Operator UX	Improved — Tune tab, Observe tab, session scripts

Primary drag: directional 30¢-band scalping with stop-losses in a market where fees and spread dominate small edges. **Arb is not the loss source** — individual arb legs show large wins/losses at settlement, but packages net positive.

Work Completed (Chronological)

Configuration Authority (31 Aug)

- Fixed governor/optimizer silently overwriting operator dashboard saves
- `canWrite / partitionPatchByAuthority` in `config/resolver.ts`
- Operator tier patches now persist; automation cannot stomp manual knobs

Sizing & Signal Stack Repair (31 Aug)

- **Sizing bug:** `maxPositionCap`: 4 from \$150-era config capped all trades at \$4; repaired to bankroll-scaled caps
- **Confidence scale:** Kelly used 0–1 assumption; real fused confidence spans ~ 0.05 – 0.46 ; added `sizingScalar()`
- **MACD units bug** (historical): signal health module added after degenerate 100% DOWN / ML-flip-to-UP incident
- **Strike forecast** (`strikeForecast.ts`): $P(\text{above strike})$ from spot, vol, time; plausibility cap at 15% edge
- **Book microstructure** (`bookMicrostructure.ts`): ladder imbalance, microprice, slippage walk
- Unit tests expanded ($\sim 364+$ passing at time of log)

Classic Paper Architecture Restore (1 Sep)

- `classicPaperStrategy()` — pre-live flow without stacked gates blocking all buys
- Disabled hold-to-settle grind, strike veto, ML on paper profile variants
- Data assurance: empty current-market set $\rightarrow$ warn not block
- Hard reset to \$1,000; governor re-enabled with DD breaker

Depth-Aware Arb & Governor (1 Sep)

- **arbDepth.ts** — walks both ask ladders; no touch-only phantom gaps
- **arbEngine.ts** — atomic packages, leg rollback on partial fill, orphan reconciliation
- Arb scans every cycle when **clobArbEnabled**; not only in arb-only mode
- Governor profiles updated to 25–38¢ band alignment

Historical Backtest Engine (1 Sep)

- **src/polyarket/backtest/** — Binance 1m → synthetic CLOB, full replay
- **6-month arb-only run:** \$10,000 → \$10,060 (+0.6%), 74 arb packages, 0 reloads
- Settlement bugfix: credit full \$1/share on arb settle
- **Caveat:** no real Polymarket CLOB history; gaps are synthetic

Observability & Operator UI (1–2 Sep)

- **observability/** — unified snapshot, alerts, SSE stream
- Dashboard tabs: **Observe** (`/#observe`), **Tune** (`/#tune`)
- Governor API: regime manual/auto, clear DD breaker
- `npm run observe`, `npm run session:200`
- Poll interval 250ms → 2000ms (SSE handles realtime)

Incident Fixes (1 Sep)

- Orphan arb leg past window end blocked all buys; scan loop hung
- `recomputeDataAssurance()` on sells; 90s scan watchdog
- Arb leg execution: fixed `!!executeTrade()` truthy-on-refusal bug (item 27)
- Arb rollback: both entry and exit fees now charged correctly

Session Work (2 Sep)

- `directionalSessionStrategy()` — 30¢ band (0.25–0.35), TP/SL exits, gates off
- Hard reset + 200-trade observer session (`directional-session-200.ts`)
- **Arb loosened:** `minArbGap` 0.5%, `arbMinMarginPct` 0.2%, `maxArbPerSlug`: 3
- **Arb UI fix:** removed fake $\pm 999\%$ TP/SL display; arb legs show “hold to settle”
- Paper reset clears governor lock + signal health state

Architecture Overview

Scan loop (every ~2s)

Markets (BTC/ETH 5m) → depth + spot + PTB

→ **Arb engine** (ladder walk, package lock) [parallel]

→ **Directional** (analyze → `alphaFusion` → `buildDecision` → Kelly)

→ Gates: edge, governor, signal health, data assurance

→ `executePendingTrade` → positions → TP/SL/settle

Governor → regime overlay (scalp / trend-ride / arb-only) + DD breaker

Figure 1: High-level decision flow

Key Modules

Module	Role
arbEngine.ts	Detect gap, lock UP+DOWN package, rollback on single-leg fill
arbDepth.ts	Fee-aware ladder walk on both asks
engines/directionalCats	Candidate scoring, edge filter, side balance
kelly.ts	Position sizing + TP/SL/hold-to-settle plan
strikeForecast.ts	Strike geometry vs spot; veto on implausible edge
signalHealth.ts	Distribution/saturation checks; suspend directional if broken
ai/governor.ts	Regime profiles, DD breaker, manual lock
edge.ts	Rolling expectancy gate (arbOnlyUntilEdge)
observability/backtest/	Snapshot, alerts, session metrics 6-month replay on synthetic books

Current State (Live Snapshot)

Captured 2 Sep 2026, 08:12 UTC from `session-observe-500-summary.json` (observer may have stopped since).

Metric	Value
Session ID	session-mtjqc931
Session started	2026-09-02 06:43 UTC
Bot running (at snapshot)	Yes
Equity	\$884.51
Peak equity (session)	\$998.70
Drawdown off peak	11.43%
Governor regime	scalp (ADX 13 range/chop)
DD breaker	Off
Price band	0.25 – 0.35 (30¢ directional)
Open positions	0
Session closes	45 (8W / 37L, 18% WR)
Session P&L	–\$3.61
Scans completed	2,713
Avg trade size	\$7.43 (range \$1.45–\$10.25)
Side balance (session)	42 UP / 3 DOWN (93% UP)
Edge gate (100 lookback)	13% WR, –\$0.22/trade expectancy
Signal health	OK, directional trustworthy
Data assurance	Score 100, canBuy true (at snapshot)
CLOB WebSocket	Connected, 8 subscriptions

Table 2: Live observability snapshot

Alerts at snapshot: stale scan loop (83s since last scan — possible hang); edge gate warning (edgeMinTrades: 999 misleading “arb-only” message while directional still allowed).

P&L Attribution (Recent Session)

Active Configuration

Arb Strategy (How It Works)

1. When $\text{ask}_{\text{UP}} + \text{ask}_{\text{DOWN}} < 1 - \text{fees} - \text{margin}$, buy equal shares of both legs
2. Walk both ask ladders — touch gap often overstates real fill

Engine	Closes	Net P&L	Notes
Directional	~26–57	~ −\$15	SL-heavy; 30¢ band
Arb (leg sum)	4 legs (2 pkgs)	+\$0.68	One leg always loses at settle
Total		~ −\$14	Directional dominates

Table 3: Engine-level attribution — arb profitable at package level

Knob	Value
minArbGap	0.5%
arbMinMarginPct	0.2%
maxArbPackages	6
maxArbPerSlug	3 (stack up to 3 arb packages per window)
arbBankrollFrac / arbMaxUsd	10% / \$45
minPrice / maxPrice	0.25 / 0.35
slPct / tpPctLow	12% / 18%
holdToSettleFavorites	false
arbOnlyUntilEdge	false
useStrikeForecast / useML	false / false
governorEnabled	true (DD breaker 15%)

3. Hold both to window settlement; one leg pays \$1/share, other \$0
4. Package net = locked edge at entry (typically \$0.25–\$0.65 per package on recent fills)
5. If one leg fails to fill, rollback the other (costs both fees)

Known Problems & Failure Modes

1. **Directional edge not demonstrated on paper.** Repeated SL hits in 30¢ band; fees ~3–7% of notional erode small targets.
2. **Gate stacking (historical).** DD breaker + edge gate + signal health + orphan blocks created long no-trade periods. Partially fixed; still complex.
3. **Live arb rarity.** Real CLOB gaps after ladder walk are infrequent; 2 arb packages in hours vs 50+ directional trades.
4. **Synthetic backtest optimism gap.** 6mo +0.6% on injected mispricing may not transfer to live microstructure.
5. **Spot vs Chainlink resolution feed.** Strike forecast can disagree with market when spot feed diverges from settlement oracle.
6. **UP-side bias (active).** Session side balance 93% UP (42 vs 3). Side-balance rule requires both outcomes passing edge filter; if DOWN rarely passes, monopoly persists.
7. **Process stability.** OOM kills (exit 137), scan hangs, orphan legs — mitigated but recurred.
8. **Hold-to-settle dead band (historical).** 0.42–0.50 neither underdog nor favorite caused unintended mid-window SL exposure.

What to Improve (Prioritized Roadmap)

P0 — Prove or Kill Directional Edge

- Run isolated **directional-only** and **arb-only** sessions (100+ closes each) with clean resets
- Measure: expectancy, fee drag, SL rate, time-in-trade, side balance

- If $30c$ band $WR < \text{break-even after fees}$ $\rightarrow$ widen band, tighten confidence gate, or pause directional
- Decision rule: no live capital until paper expectancy > 0 over 200 closes in chosen profile

P1 — Real Book Data for Arb

- Record CLOB WS snapshots to disk (`data/clob-snapshots/`)
- Replay recorded books in backtest instead of synthetic `bookSim` gaps
- Calibrate `minArbGap` / `arbMinMarginPct` on real gap distribution
- Target: know live arb hit rate (packages/day) before sizing up

P1 — Fee-Adjusted Expectancy Dashboard

- Package-level arb P&L card (not per-leg) in UI
- Per-engine rolling expectancy, break-even WR , Kelly validity
- Alert when directional WR below break-even for 20+ closes

P2 — Resolution Feed Alignment

- Source spot from same feed Polymarket uses for PTB resolution (Chainlink stream)
- Reduce false strike-forecast vetoes and wrong-side entries on feed skew

P2 — Simplify Gate Stack

- Single “can trade directional / can trade arb” panel with explicit reasons
- Preset profiles: `arb-hunt`, `directional-30c`, `classic-paper` — one-click, no conflicting overlays
- Edge gate: disable or raise `edgeMinTrades` on paper (currently misleading “arb-only” alerts)

P3 — Reliability

- Memory profiling / OOM guardrails on scan loop
- Automated orphan detection + unwind cron
- Health endpoint: last scan age, `_scanning` lock state

P3 — Live Readiness

- CTF instant merge path tested on testnet
- Live safety caps enforced (`assertLiveSafetyCaps`)
- Arb-only live pilot with \$50 cap before directional live

Recommended Next Session

1. **Stop blended session.** Directional is bleeding; arb is too rare to offset.
2. **Fork A (arb focus):** `forceArbOnly: true`, loosened gaps (current), 48h observe — count packages/day and net P&L.
3. **Fork B (directional refine):** Widen to 0.22–0.38, raise `minConfidence`, require book quality > 0.25 , 100-trade clean run.
4. **Do not** scale position size until rolling expectancy turns positive.
5. Export this report + `session-observe-500.jsonl` weekly for audit trail.

Appendix: Commands & Paths

```
Project root:      /home/david/Zinger
Dashboard:         http://127.0.0.1:3000  (#observe, #tune)
Start bot:         npm start
200-trade session: npm run session:200
Observer:          npm run observe -- --target=200
6mo backtest:      npx tsx scripts/historical-backtest.ts --months=6 --arb-only
Compile report:    cd docs/reports && tectonic zinger-status-report.tex
Dev log:           DEVELOPMENT.md
Observer log:      data/session-observe-500.jsonl
Observer summary:  data/session-observe-500-summary.json
Backtest result:   data/backtest/backtest-6mo-arb-2026-09-01T17-27-57.json
PDF output:        docs/reports/zinger-status-report.pdf
```